
HI-LO GUESS GAME

A Number Puzzle Game - (DOM API + Events)

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Lab Introduction

Prerequisites

DOM / Events [Lectures] . Software Requirements: Chrome browser, any code editor/IDE.

Motivation

Learn to use Document Object Model API & JavaScript to make interactive HTML-based apps.

Goal

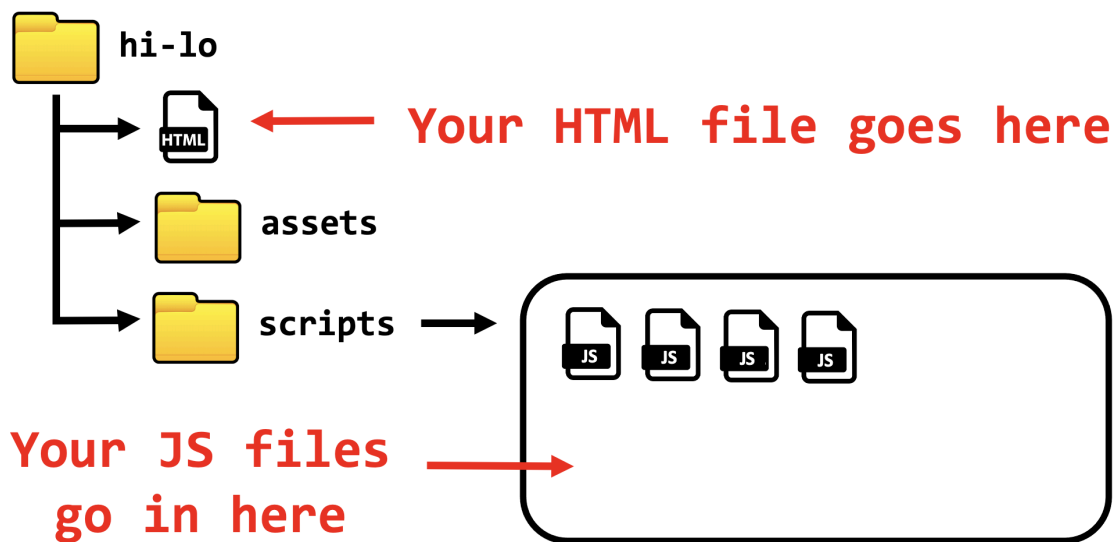
Build a compelling, fun version of the Hi-Lo Game that runs in Browser using JavaScript & DOM API.

Learning Objectives

- Document Object Model API to access & update HTML
- Event-driven System design
- Agile practices: Minimal Viable Product
- Model-View-Controller Design Pattern
- UI/UX considerations/improvements
- Timed Events and Date object

Project Architecture:

Start this project by downloading the starter files from github. See the project structure below.



Download Starter files:

<https://github.com/scalemailed/hilo-js-dom/archive/master.zip>

Document Object Model (DOM)

The DOM gives JavaScript access to the HTML document. The DOM is a built-in object with functions for manipulating the HTML. The **DOM converts HTML elements into JavaScript objects.**

- **document object:** The global object that is the entry point to the DOM API.
 - **element objects:** The document object accesses element objects via ***getElementById(id)***. These element objects are the JS models of the HTML elements from the web page.
 - **innerHTML:** A property of element objects that maintains its HTML in markup-notated strings. By changing the value of the innerHTML, the element may add, change, delete HTML from DOM.
-

Browser (HTML) Events

Web browsers are event-driven systems. A browser generates & manages events based on certain actions or triggers. Those events may be used by JS apps to trigger actions or behaviors.

- **Event listener:** An event listener registers an element object to listen to the event queue for specific types of event. When that event occurs it invokes a callback function
 - **Callback function:** a function within the JS app that is passed as a reference to the event system. When a event occurs the event system invokes the callback function
 - **Event object:** Whenever a callback function is invoked, the browser passes an Event object as a parameter. The Event object has properties specific to that event.
 - **Event type:** Every event object has a type. The event listener matches the Event objects type to a callback function, similar to a key-value pair. Event types are strings such as "click" or "load"
-

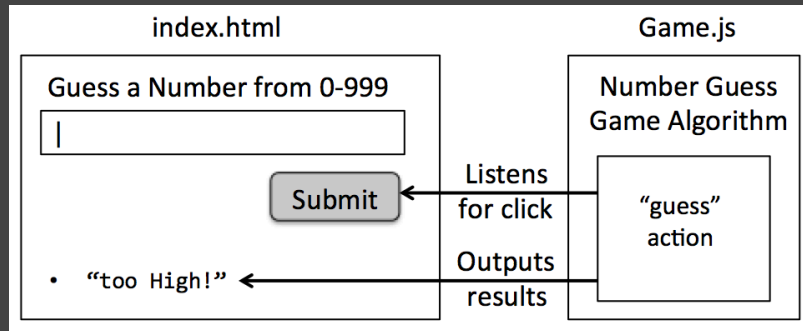
Model-View-Controller (MVC)

MVC is a design pattern where an app's responsibilities are divided into Model, Controller and View.

- **Model:** The model manages the app's core logic without concerns for handling input or output.
- **Controller:** The controller handles user input to send data & invoke actions within the model
- **View:** The view manages outputting data from the model & presenting it to the user.

Goal 0: Single Page App - Iterative Builds

Part 1: Minimal Viable Product (MVP) → Focus development cycle on app's necessities.



Part 2: Refactor App with Model-View-Controller Pattern & Improve UI

There is a number between 000-999.

Number of attempts left:

Enter a Guess:

+	+	+
3	2	5
-	-	-
Submit		

- 500 is too high
- 250 is too low

This version refactors codebase into 3 parts:

- model: manages app logic
- view: manages app's output
- controller: manages app's inputs

This version also ensures that only valid inputs are allowed and that all devices are supported.

Part 3: Refactor App into Time-based Challenge with Graphics



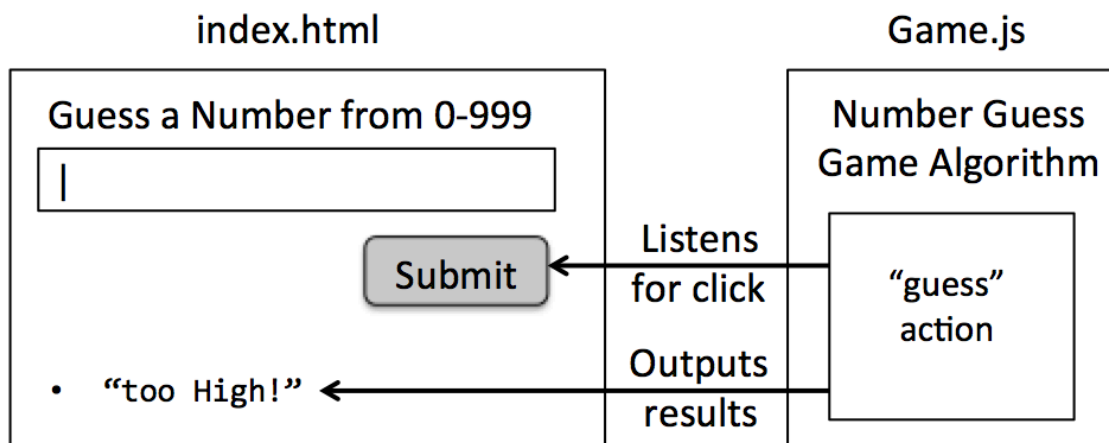
This version refactors codebase to improve UX.

From a Turn-based game into a Time-based for a novel & compelling version of Hi-Lo. Try to hack the system before the clock runs out!

Use graphics instead of HTML inputs

PART 1:

Turn-based, Text-based with M.V.P. design



Goal 1-0: Minimal Viable Product (MVP)

Summary:

This lab uses an agile approach for developing the Hi-Lo game, whereby we'll focus the initial build on delivering a Minimal Viable Product (MVP). A MVP is a version of an app with just enough features to be usable by early users who can then evaluate it for future development.

MVP Specifications:

The specification defines the necessary features for this version of the application

- Capture input from user through browser's viewport
- Update the browser's viewport with the player's results
- Implement the core game logic for executing the Hi-Lo game

Goal 1-1: HTML for Hi-LO Game

'APPROACH' → PLAN PHASE

Make an HTML document that gives the game's instructions & its user inputs

'APPLY' → DO PHASE

Step 1: Create an index.html file & implement it with the base HTML content.

index.html

```
<html>
<head></head>
<body>
  <p>There is a number between 000-999.</p>
  <p>You have 10 guesses.</p>
  <span>Enter a Guess:</span>
  <input type="text">
  <input type="button" value="Submit">
</body>
</html>
```

'APPROVE' → TEST PHASE

Open the index.html document in the browser and ensure that it displays

There is a number between 000-999.

You have 10 guesses.

Enter a Guess:

Goal 1-2: DOM for JS controller (Event Listener)

'APPROACH' → PLAN PHASE

In this iteration, the HTML inputs are accessed from JavaScript which "listens" for a button click and triggers a "callback" function in response.

'APPLY' → DO PHASE

Step 1: Link the game.js file to index.html & give its inputs id attributes to access them from JS

index.html

```
<html>
  <head></head>
  <body>
    <p>There is a number between 000-999.</p>
    <p>You have 10 guesses.</p>
    <span>Enter a Guess:</span>
    <input type="text" id="guess-text">
    <input type="button" value="Submit" id="guess-button">
    <script src="scripts/game.js"></script>
  </body>
</html>
```

Step 2: Use document to get html by id & add an event listener to button with callback function

game.js

```
/* Get HTML Elements as JS objects */
const button = document.getElementById("guess-button");
const number = document.getElementById("guess-text");

/* Add Event Listener to button with callback function*/
button.addEventListener("click", guessNumber);

/* Callback function for event: Button click */
function guessNumber() {
  const guess = number.value;
  console.log(guess);
}
```

'APPROVE' → TEST PHASE

There is a number between 000-999.

You have 10 guesses.

Enter a Guess:

in the console:

500

Goal 1-3: Hi-Lo Game logic in JS

'APPROACH' → PLAN PHASE

Implement the basic algorithm for the Hi-Lo game. The game's output displays to the console.

'APPLY' → DO PHASE

Step 1: Create variables for the random passcode (0-999) & the number of tries remaining.

game.js

```
/* Hi-Lo Game Data*/  
const passcode = Math.floor( Math.random() * 1000 );  
let tries = 10;
```

Step 2: Implement the logic for evaluating a guess to determine if user won, lost or gets a clues

game.js

```
/* Callback function for event: Button click */  
function guessNumber() {  
  const guess = number.value;  
  tries--;  
  console.log(`Number of attempts left: ${tries}`);  
  if ( guess == passcode){  
    console.log(`You win! Got it in ${10-tries} attempts`);  
  }  
  else if (tries < 0){  
    console.log(`You lose! The passcode was ${passcode}`);  
  }  
  else{  
    giveClue(guess)  
  }  
}
```

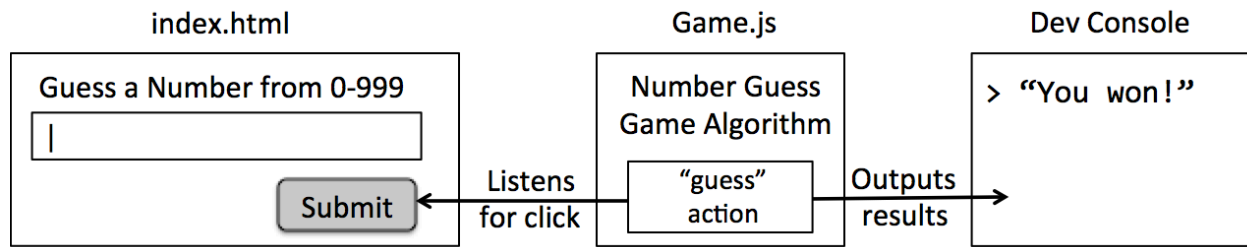
Step 3: Define a function that gives a clue, i.e. whether a guess was too low or too high.

game.js

```
/*Give Clue */  
function giveClue(guess){  
  if (guess > passcode){  
    console.log(`${guess} is too High!`);  
  }  
  else{  
    console.log(`${guess} is too Low!`);  
  }  
}
```

'APPROVE' → TEST PHASE

Play the game. Submit your guesses using the button & read result from the console.



Goal 1-4: DOM for JS view (innerHTML)

'APPROACH' → PLAN PHASE

Use DOM API to update the HTML elements from the JavaScript

'APPLY' → DO PHASE

Step 1: Add an empty list for clues, with an id, also add an id to the paragraph for attempts

index.html → *<body>*

```
<body>
  <p>There is a number between 000-999.</p>
  <p id='attempts'>You have 10 guesses.</p>
  <span>Enter a Guess:</span>
  <input type="text" id="guess-text">
  <input type="button" id="guess-button" value="Submit">
  <ul id='clues'></ul>
  <script src="scripts/game.js"></script>
</body>
```

Step 2: Use the document object to access the HTML elements as JavaScript objects.

game.js

```
const attemptsView = document.getElementById("attempts");
const cluesView = document.getElementById("clues");
```

Step 3: Overwrite innerHTML of attempts text after a guess, & overwrite the body on gameover

game.js

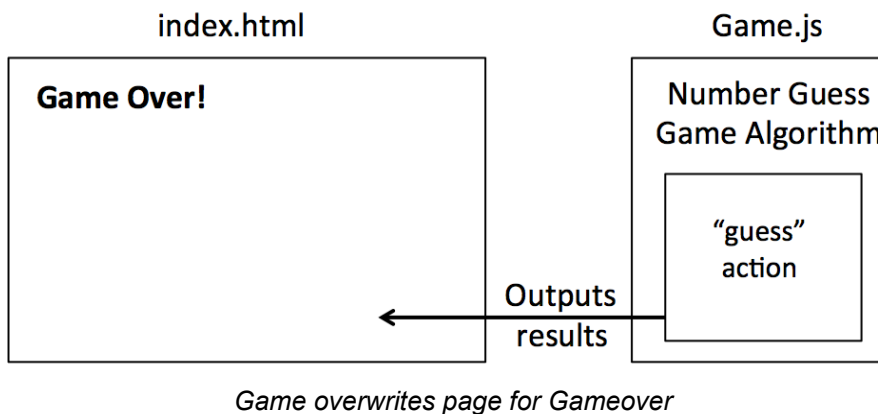
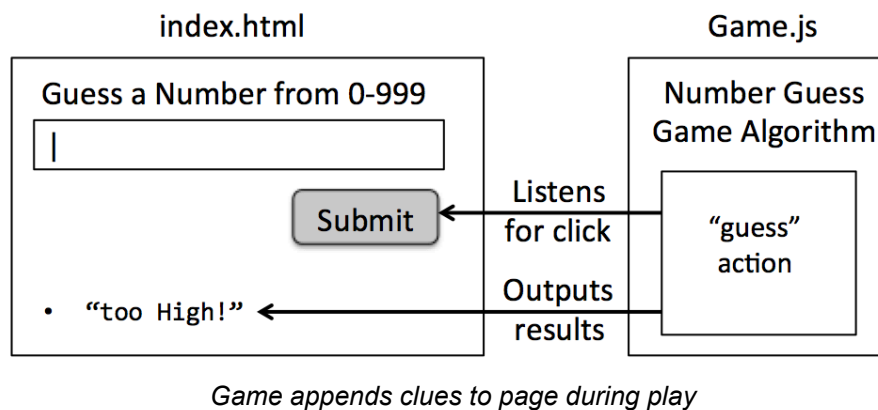
```
/* Callback function for event: Button click */
function guessNumber() {
  const guess = number.value;
  tries--;
  attemptsView.innerHTML = `Number of attempts left: ${tries}`;
  if ( guess == passcode){
    document.body.innerHTML = `<h1>You win!</h1> <p>Got it in ${10-tries} attempts</p>`;
  }
  else if (tries < 0){
    document.body.innerHTML = `<h1>You lose!</h1> <p>The passcode was ${passcode}</p>`;
  }
  else{
    giveClue(guess)
  }
}
```

Step 4: Append to the innerHTML of the clues list with a new clue in a list item.

game.js

```
/*Give Clue */  
function giveClue(guess){  
  if (guess > passcode){  
    cluesView.innerHTML += `<li>${guess} is too High!</li>`;  
  }  
  else{  
    cluesView.innerHTML += `<li>${guess} is too Low!</li>`;  
  }  
}
```

'APPROVE' → TEST PHASE



PART 2:

Turn-based, Text-based with MVC pattern

There is a number between 000-999.

Number of attempts left:

Enter a Guess:

+	+	+
3	2	5
-	-	-
Submit		

- 500 is too high
- 250 is too low

Goal 2-0: MVC Design Pattern

Summary:

Let's create a new mockup for an improved version of the Puzzle Game. Each iterative goal will revolve around adding in those features.

Countdown Challenge Game Mockup

There is a number between 000-999.

Number of attempts left:

Enter a Guess:

+	+	+
3	2	5
-	-	-

Submit

- 500 is too high
- 250 is too low

Specifications:

- *Improved Code Maintainability:* Adopt better software engineering principles and refactor code into three responsibilities: **model, views, controllers**.
- *Improved Controls:* Fault-tolerant user inputs that prevent invalid values. Ensure that inputs do not require a keyboard, but also support mobile users.
- *Improved Views:* Better UI/UX with more graphics and styling

Responsibilities across M-V-C App

View	Manages the output from model to user
Controller	Manages the input from user into model
Model	The program's logic (game rules only, no input/output concerns)

Goal 2-1: Refactor with MVC Pattern

'APPROACH' → PLAN PHASE

Model-View-Controller (MVC) Architecture:

Divide code for Hi-Lo game into three files

	model.js	view.js	controller.js
variables	passcode tries	-	-
functions	guessNumber() giveClue()	printAttemptsRemaining() printClue() printGameOver()	initControls() buttonEvent()

'APPLY' → DO PHASE

Step 0: In scripts folder, make 3 new files named: model.js, controllers.js, views.js

Step 1: [JS] Implement the Model → Hi-Lo game logic

model.js

```
const passcode = Math.floor( Math.random()*1000 );
let tries = 10;

function guessNumber(guess){
  tries--;
  if ( guess == passcode ){
    printGameOver('WIN');
  }
  else if (tries <= 0){
    printGameOver('LOSE');
  }
  else{
    printAttemptsRemaining(tries);
    giveClue(guess);
  }
}

function giveClue(guess){
  if (guess > passcode){
    printClue('HI', guess);
  }
  else{
    printClue('LO', guess);
  }
}
```


Step 2: [JS] Implement the Controller → Inputs for Hi-Lo

controllers.js

```
initControls();

function initControls(){
  const button = document.getElementById("guess-button");
  button.addEventListener("click", buttonEvent);
}

function buttonEvent(){
  const number = document.getElementById("guess-text");
  guessNumber(number.value)
}
```

Step 3: [JS] Implement the View → Outputs for Hi-Lo

views.js

```
function printAttemptsRemaining(tries){
  const attemptsText = document.getElementById("attempts");
  attemptsText.innerHTML = `Number of attempts left: ${tries}`;
}

function printClue(status, guess){
  const clueText = document.getElementById("clues");
  const clue = (status === 'HI') ? `<li>${guess} is too high</li>` : `<li>${guess} is too low</li>`;
  clueText.innerHTML += clue;
}

function printGameOver(status){
  if (status === 'WIN'){
    var message = `<h1>You Win!</h1> <p>Got it in ${10-tries} tries.</p>`;
  }
  else{
    var message = `<h1>You Lose!</h1> <p>The number was: ${passcode}</p>`;
  }
  document.body.innerHTML = message;
}
```

Step 4: [HTML] Update the index.html file's script tags to import the three new JS files.

index.html → *<body>*

```
<body>
  <p>There is a number between 000-999.</p>
  <p id="attempts">Number of attempts left: 10</p>
  <span>Enter a Guess:</span>
  <input type="text" id="guess-text">
  <input type="button" id="guess-button" value='Submit'>
  <ul id="clues"></ul>
  <script src="scripts/views.js"></script>
  <script src="scripts/model.js"></script>
  <script src="scripts/controllers.js"></script>
</body>
```

'APPROVE' → TEST PHASE

Play the game! Make sure it still works as it did before the refactoring.

Goal 2-2: Refactor User Inputs

'APPROACH' → PLAN PHASE

New UI that improves precision & ensures valid input. Scheme is based on a combination lock.



Combination Lock

There is a number between 000-999.

Number of attempts left: 10

Enter a Guess:

Submit

New UI

'APPLY' → DO PHASE

Step 1: [HTML] Refactor the text input tags from one to three number fields.

index.html → *<body>*

```
<body>
  <p>There is a number between 000-999.</p>
  <p id="attempts">Number of attempts left: 10</p>
  <span>Enter a Guess:</span>
  <div>
    <input type="number" min="0" max="9" value="0" id="digit-100s">
    <input type="number" min="0" max="9" value="0" id="digit-10s">
    <input type="number" min="0" max="9" value="0" id="digit-1s">
  </div>
  <input type="button" value='Submit' id="guess-button">
  <ul id="clues"></ul>

  <script src="scripts/views.js"></script>
  <script src="scripts/model.js"></script>
  <script src="scripts/controllers.js"></script>
</body>
```

Step 2: [JS] Update the buttonEvent function to get the guess

controllers.js → *buttonEvent()*

```
function buttonEvent(){
  const hundreds = document.getElementById("digit-100s");
  const tens = document.getElementById("digit-10s");
  const ones = document.getElementById("digit-1s");
  const number = "" + hundreds.value + tens.value + ones.value;
  guessNumber(number);
}
```

'APPROVE' → TEST PHASE

Play the game! Compatible with mouse/keyboard. Note: keyboards may input invalid numbers. Doesn't work with touch devices. Improve UI to support touch and prevent invalid values from keyboard

Goal 2-3: Define a class Guess

'APPROACH' → PLAN PHASE

Model guess as a class maintained by the model that the view/controller references

'APPLY' → DO PHASE

Step 1: [HTML] Buttons to increment guess (*Controller*) & Read-only field to display it (*View*)

index.html → `<body>`

```
<body>
  <p>There is a number between 000-999.</p>
  <p id="attempts">Number of attempts left: 10</p>
  <span>Enter a Guess:</span>

  <div>
    <input type="button" id="up-100s" value=' + '>
    <input type="button" id="up-10s" value=' + '>
    <input type="button" id="up-1s" value=' + '>
  </div>
  <div>
    <input disabled type="number" min="0" max="9" value="0" id="digit-100s">
    <input disabled type="number" min="0" max="9" value="0" id="digit-10s">
    <input disabled type="number" min="0" max="9" value="0" id="digit-1s">
  </div>
  <input type="button" value='Submit' id="guess-button">

  <ul id="clue"></ul>

  <script src="scripts/Guess.js"></script>
  <script src="scripts/views.js"></script>
  <script src="scripts/model.js"></script>
  <script src="scripts/controllers.js"></script>
</body>
```

Step 2: [JS] Create a Guess class that models the attributes & behaviors of a guess object

Guess.js

```
class Guess{
  constructor(){
    this.hundreds = 0;
    this.tens = 0;
    this.ones = 0;
  }
  toString(){
    return "" + this.hundreds + this.tens + this.ones;
  }
  increment(key){
    this[key] = (this[key] + 1) % 10;
  }
}
```

Note: Objects are Dictionaries, so can access any attribute given its name as key in brackets.

Step 3: [JS] Instantiate an instance of Guess in the model.js

model.js → global variables

```
const passcode = Math.floor( Math.random()*1000 );
let tries = 10;
const guess = new Guess();
```

Step 4: [JS] Add a function to print the state of guess with DOM in view.js

views.js

```
function printDigits(){
  document.getElementById("digit-100s").value = guess.hundreds;
  document.getElementById("digit-10s").value = guess.tens;
  document.getElementById("digit-1s").value = guess.ones;
}
```

Step 5: [JS] Add a callback function (controller) to increment the guess and print it to the view

controllers.js

```
function incrementEvent(key){
  guess.increment(key);
  printDigits();
}
```

Step 6: [JS] Add event listeners to the '+' buttons that trigger the callback function to increment

controllers.js

```
function initControls(){
  const button = document.getElementById("guess-button");
  button.addEventListener("click", buttonEvent);
  const up100s = document.getElementById("up-100s");
  const up10s = document.getElementById("up-10s");
  const up1s = document.getElementById("up-1s");
  up100s.addEventListener("click", () => incrementEvent('hundreds'));
  up10s.addEventListener("click", () => incrementEvent('tens'));
  up1s.addEventListener("click", () => incrementEvent('ones'));
}
```

Step 7: [JS] Get the state of guess from the instance and send to the game logic on a button event

controllers.js

```
function buttonEvent(){
  const number = guess.toString();
  guessNumber(number)
}
```

'APPROVE' → TEST PHASE

Play the game! This iteration now works on all devices and prevents invalid states. However, you can only increment the value up which doesn't offer a good UX.

Goal 2-4: Decrement Controls

'APPROACH' → PLAN PHASE

Improve User Experience (UX) by adding a decrement control into the game.

'APPLY' → DO PHASE

Step 1: [HTML] Create HTML buttons for decrementing.

index.html → <body>

```
<body>
  <p>There is a number between 000-999.</p>
  <p id="attempts">Number of attempts left: 10</p>
  <span>Enter a Guess:</span>
  <div>
    <input type="button" id="up-100s" value=' + '>
    <input type="button" id="up-10s" value=' + '>
    <input type="button" id="up-1s" value=' + '>
  </div>
  <div>
    <input disabled type="number" min="0" max="9" value="0" id="digit-100s">
    <input disabled type="number" min="0" max="9" value="0" id="digit-10s">
    <input disabled type="number" min="0" max="9" value="0" id="digit-1s">
  </div>
  <div>
    <input type="button" id="down-100s" value=' - '>
    <input type="button" id="down-10s" value=' - '>
    <input type="button" id="down-1s" value=' - '>
  </div>
  <input type="button" value='Submit' id="guess-button">
  <ul id="clues"></ul>
  <script src="scripts/Guess.js"></script>
  <script src="scripts/views.js"></script>
  <script src="scripts/model.js"></script>
  <script src="scripts/controllers.js"></script>
</body>
```

Step 2: [JS] Add a decrement method into the body of the Guess class

Guess.js

```
decrement(key){
  this[key] = (this[key] > 0) ? this[key]-1 : 9;
}
```

Note: Objects are Dictionaries, so can access any attribute given its name as key in brackets.

Step 3: [JS] Add a callback function in controllers to decrement guess and print its value

controllers.js

```
function decrementEvent(key){
  guess.decrement(key);
  printDigits();
}
```

Step 4: [JS] Add event listeners to the down buttons that trigger the callback function to decrement

controllers.js

```
function initControls(){
  const button = document.getElementById("guess-button");
  button.addEventListener("click", buttonEvent);

  const up100s = document.getElementById("up-100s");
  const up10s = document.getElementById("up-10s");
  const up1s = document.getElementById("up-1s");
  up100s.addEventListener("click", () => incrementEvent('hundreds'));
  up10s.addEventListener("click", () => incrementEvent('tens'));
  up1s.addEventListener("click", () => incrementEvent('ones'));

  const down100s = document.getElementById("down-100s");
  const down10s = document.getElementById("down-10s");
  const down1s = document.getElementById("down-1s");
  down100s.addEventListener("click", () => decrementEvent('hundreds'));
  down10s.addEventListener("click", () => decrementEvent('tens'));
  down1s.addEventListener("click", () => decrementEvent('ones'));
}
```

'APPROVE' → TEST PHASE

Play the game. This iteration now works on all devices and prevents invalid states. This version offers superior UX with increments/decrements & rollovers. This game may still be improved with graphics & time events.

PART 3:

Time-based, Graphics-based with MVC pattern

Passcode between: 000-999.

Time left: 10

Enter the passcode:

▲	▲	▲
0	0	0
▼	▼	▼

ENTER

Goal 3-0: Real-time Design

Summary:

Let's create a new mockup for an improved version of the Puzzle Game. Each iterative goal will revolve around adding in those features.

Countdown Challenge Game Mockup

Passcode between: 000-999.

Time left: 10

Enter the passcode:

▲	▲	▲
0	0	0
▼	▼	▼

ENTER

The mockup shows a game interface for a 'Countdown Challenge'. At the top, it says 'Passcode between: 000-999.' Below that, a timer displays 'Time left: 10'. The instruction 'Enter the passcode:' is followed by a 3x3 grid of buttons. The top row has three upward-pointing red triangles. The middle row has three buttons, each displaying a red digital '0'. The bottom row has three downward-pointing red triangles. Below the grid is a large green button with the word 'ENTER' in white capital letters.

Specifications:

- *Improved Gameplay:* Design a more compelling game experience by forcing the player into inputting choices quicker, where attempts are limited by a timer instead of a count.
- *Improved Views:* Replace all HTML elements with art & animated graphics.

Goal 3-1: Graphical Controllers

'APPROACH' → PLAN PHASE

Replace the HTML button elements with rendered graphics as the controllers

'APPLY' → DO PHASE

Step 1 [HTML] Replace the buttons tags and input tags with image tags.

index.html → *<body>*

```
<body>
  <p>There is a number between 000-999.</p>
  <p id="attempts">Number of attempts left:</p>
  <span>Enter a Guess:</span>

  <div>
    
    
    
  </div>
  <div>
    <input disabled type="number" min="0" max="9" value="0" id="digit-100s">
    <input disabled type="number" min="0" max="9" value="0" id="digit-10s">
    <input disabled type="number" min="0" max="9" value="0" id="digit-1s">
  </div>
  <div>
    
    
    
  </div>
  <div>
    
  </div>

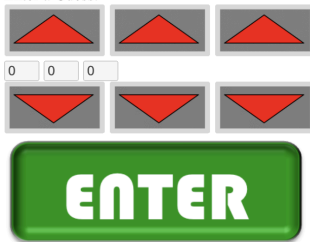
  <ul id="clue"></ul>

  <script src="scripts/Guess.js"></script>
  <script src="scripts/views.js"></script>
  <script src="scripts/model.js"></script>
  <script src="scripts/controllers.js"></script>
</body>
```

There is a number between 000-999.

Number of attempts left:

Enter a Guess:



'APPROVE' → TEST PHASE

Play the game! It now has graphical animated buttons!

- The view is ugly disabled HTML inputs at this point.

Goal 3-2: Graphical View - Digits

'APPROACH' → PLAN PHASE

Render the view graphically: The Guess's digits.

'APPLY' → DO PHASE

Step 1: [HTML] Replace the input tags with image tags.

index.html → `<body>`

```
<body>
  <p>There is a number between 000-999.</p>
  <p id="attempts">Number of attempts left:</p>
  <span>Enter a Guess:</span>

  <div>
    
    
    
  </div>
  <div>
    
    
    
  </div>
  <div>
    
    
    
  </div>
  <div>
    
  </div>

  <ul id="clues"></ul>

  <script src="scripts/Guess.js"></script>
  <script src="scripts/views.js"></script>
  <script src="scripts/model.js"></script>
  <script src="scripts/controllers.js"></script>
</body>
```

Step 2: [JS] Refactor the printDisplay function to update the src attribute for the image.

views.js

```
function printDigits(){
  document.getElementById("digit-100s").src = `assets/${guess.hundreds}.png`;
  document.getElementById("digit-10s").src = `assets/${guess.tens}.png`;
  document.getElementById("digit-1s").src = `assets/${guess.ones}.png`;
}
```

'APPROVE' → TEST PHASE

There is a number between 000-999.

Number of attempts left:

Enter a Guess:

ENTER

Play the game!

It now has graphical controllers & views.

- However the clues are still an ugly HTML list.

Goal 3-3: Graphical View - Clue

'APPROACH' → PLAN PHASE

Render the view graphically: Clues.

'APPLY' → DO PHASE

Step 1: [HTML] Remove the `<ul id='clues'></ul>` element

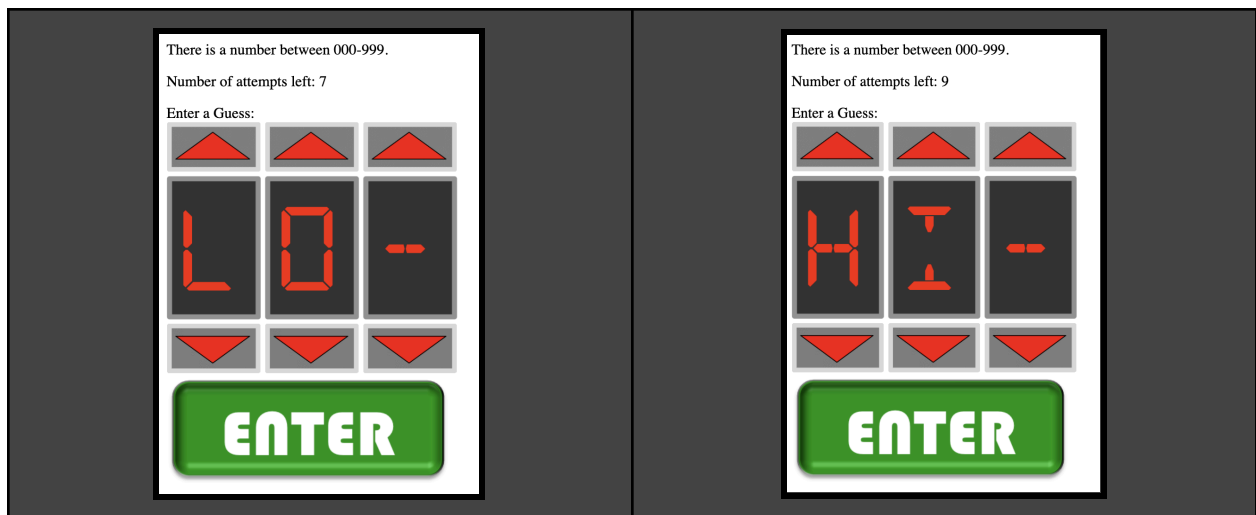
Step 2: [JS] Refactor the `printClue()` function to update the `src` attribute of the image. Destructuring is used in order to assign the three individual variables.

views.js

```
function printClue(status, guess){  
  const [digit100,digit10,digit1] = status=='HI' ? ['H','I','-'] : ['L','O','-'];  
  document.getElementById("digit-100s").src = `assets/${digit100}.png`;   
  document.getElementById("digit-10s").src = `assets/${digit10}.png`;   
  document.getElementById("digit-1s").src = `assets/${digit1}.png`;   
}
```

'APPROVE' → TEST PHASE

Play the game! The game has graphic controllers & graphical views for the digits and clues. However, the clue obfuscates the view of the digits until the player increments/decrements the guess. This is poor UI/UX!



Goal 3-4: Time Event: Overwrite View

'APPROACH' → PLAN PHASE

(Goal): Create a timer that overwrites the clue-view back into digits-view after 1 second.

To measure elapses in time, continually invoke a function & check the difference between 'now' time & 'then' time, if the duration is met, then perform the action & reset the 'now' time.

'APPLY' → DO PHASE

Step 1: [JS] Use Date class to get the current time in epoch time (*number of milliseconds since 1970*)

model.js → global variables

```
const passcode = Math.floor( Math.random()*1000 );
let tries = 10;
const guess = new Guess();
let then = Date.now();
```

Step 2: [JS] Define a main looping function that prints the digits after a second has passed. The loop recursively occurs via the requestAnimationFrame function.

model.js

```
function main(){
  const now = Date.now();
  if (now - then > 1000){
    printDigits();
  }
  requestAnimationFrame(main);
}
main();
```

Step 3: [JS] Update the 'then' time everytime the clue is displayed to start the 1000ms countdown.

views.js

```
function printClue(status, guess){
  const [digit100,digit10,digit1] = status=='HI' ? ['H','I','-'] : ['L','O','-'];
  document.getElementById("digit-100s").src = `assets/${digit100}.png`;
  document.getElementById("digit-10s").src = `assets/${digit10}.png`;
  document.getElementById("digit-1s").src = `assets/${digit1}.png`;
  then = Date.now();
}
```

'APPROVE' → TEST PHASE

Play the game. The clue view should display for only a second before refreshing back to the digit view

Goal 3-5: Time Event: Turn-based to Time-based

'APPROACH' → PLAN PHASE

(Timer logic) Use a timer instead of a counter for managing game over conditions. To manage game state over time, use a variable that tracks if the game is over or not.

'APPLY' → DO PHASE

Step 1: [JS] Initialize a timeleft variable to 30 & Initialize a gameover variable to false

model.js → global variables

```
const passcode = Math.floor( Math.random()*1000 );
let tries = 10;
const guess = new Guess();
let then = Date.now();
let timeLeft = 30;
let gameover = false;
```

Step 2: [JS] Update gameover variable in win condition of guessNumber

model.js

```
function guessNumber(guess){
  tries--;
  if ( guess == passcode ){
    gameover = true;
    printGameOver('WIN');
  }
  else{
    giveClue(guess);
  }
}
```

Step 3: [JS] Update view to report in time left instead of attempts remaining

views.js

```
function printAttemptsRemaining(tries){
  const attemptsText = document.getElementById("attempts");
  attemptsText.innerHTML = `<h2>Time left: ${timeLeft}</h2>`;
}
```

Step 4: [JS] Update view to report gameover win with `timeleft` instead of `tries left`

views.js

```
function printGameOver(status){
  if (status === 'WIN'){
    var message = `

# You Win!

 <p>Got it in ${30-timeLeft} seconds.</p>` ;
  }
  else{
    var message = `

# You Lose!

 <p>The number was: ${passcode}</p>`;
  }
  document.body.innerHTML = message;
}
```

Step 5: [JS] Update main method to refresh the time left view after every second

model.js

```
function main(){
  const now = Date.now();
  if (gameover){
    return;
  }
  else if (timeLeft <= 0){
    printGameOver('LOSE');
  }
  else if (now - then > 1000){
    timeLeft--;
    printDigits();
    printAttemptsRemaining();
    then = Date.now();
  }
  requestAnimationFrame(main);
}
```

'APPROVE' → TEST PHASE

Play the game! The game should now be time-based instead of turn-based. Fin.

Concluding Notes

Graphics-view approach

Since the app was divided into MVC architecture, converting to a graphics view was trivial. Note the strategy we employed to programmatically manage which image asset to load.

Future Improvements

- Improve the style of the game with CSS such as background colors and centering
- Store fastest time in leaderboard & save in local storage.
- Add a difficulty mode that supports either larger passcodes or different number bases
 - For the number of guesses to solve use: ceiling of $\log_2(\text{maxValue})$.

Lab Submission

Compress your project folder into a zip file and submit on Moodle.